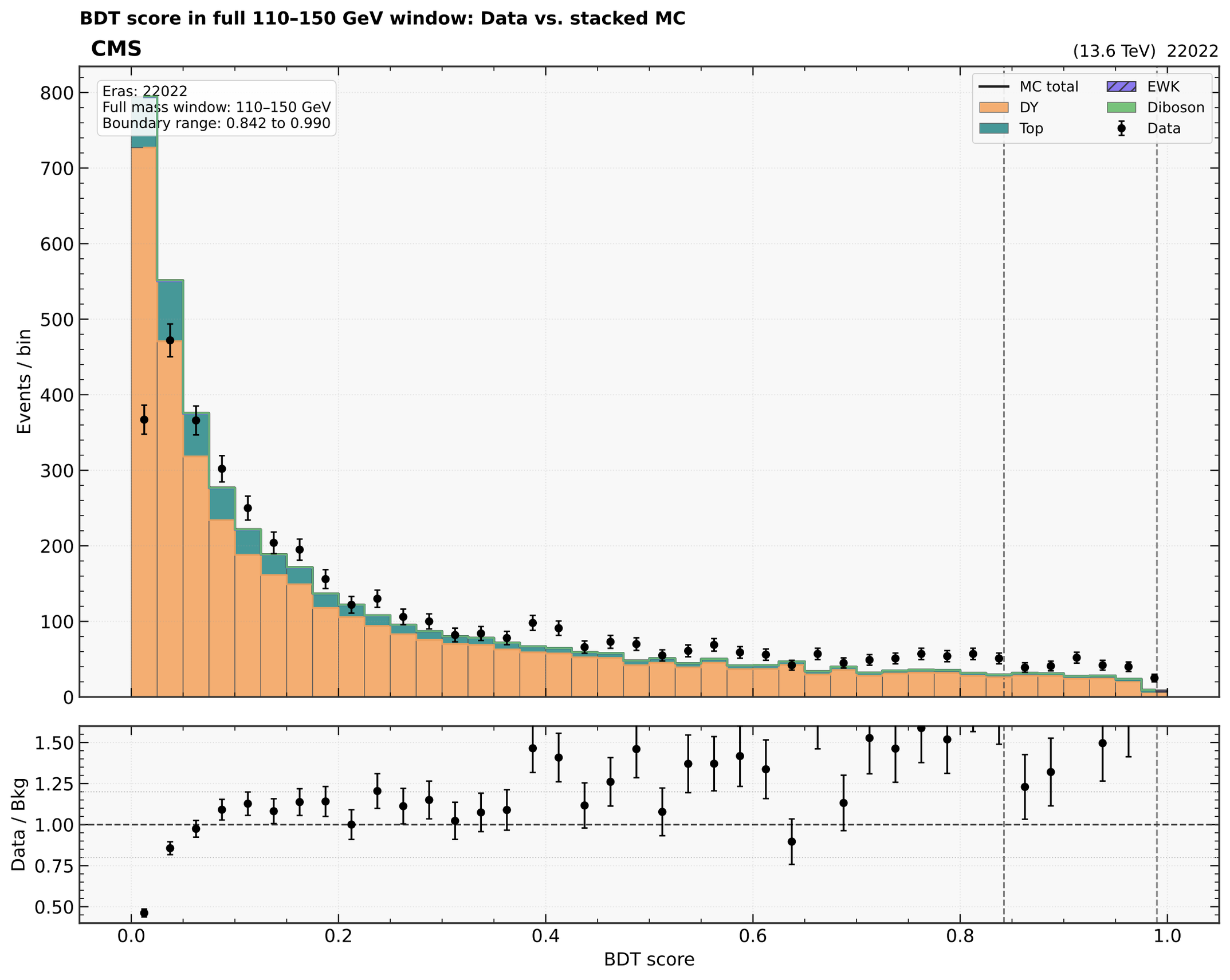
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 22022
Category edges : 0.0000, 0.8422, 0.9712, 0.9899, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ^2/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_{CB} vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

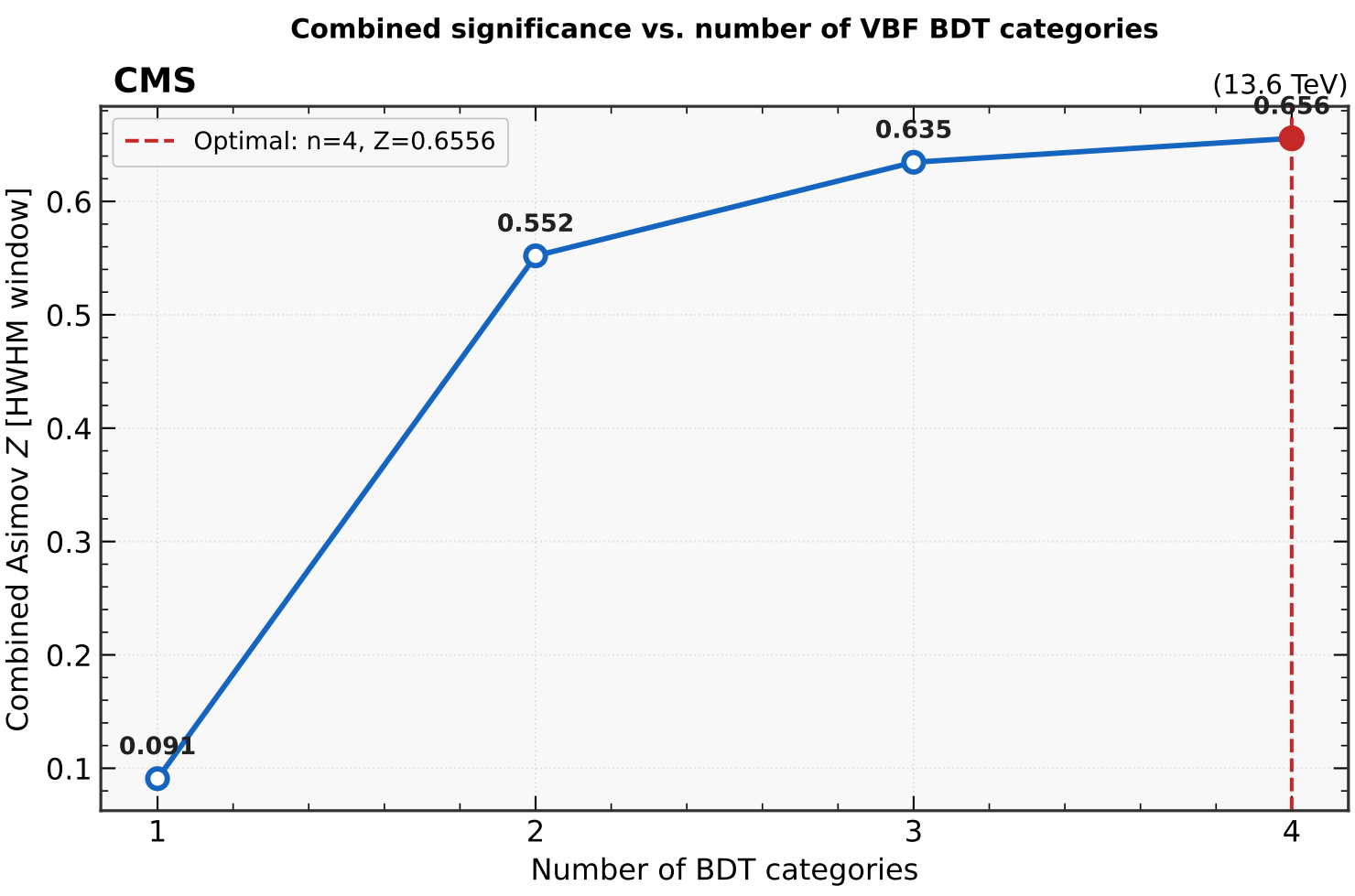

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 22022
Eras: 22022
Full mass window: 110–150 GeV
Boundary range: 0.842 to 0.990
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

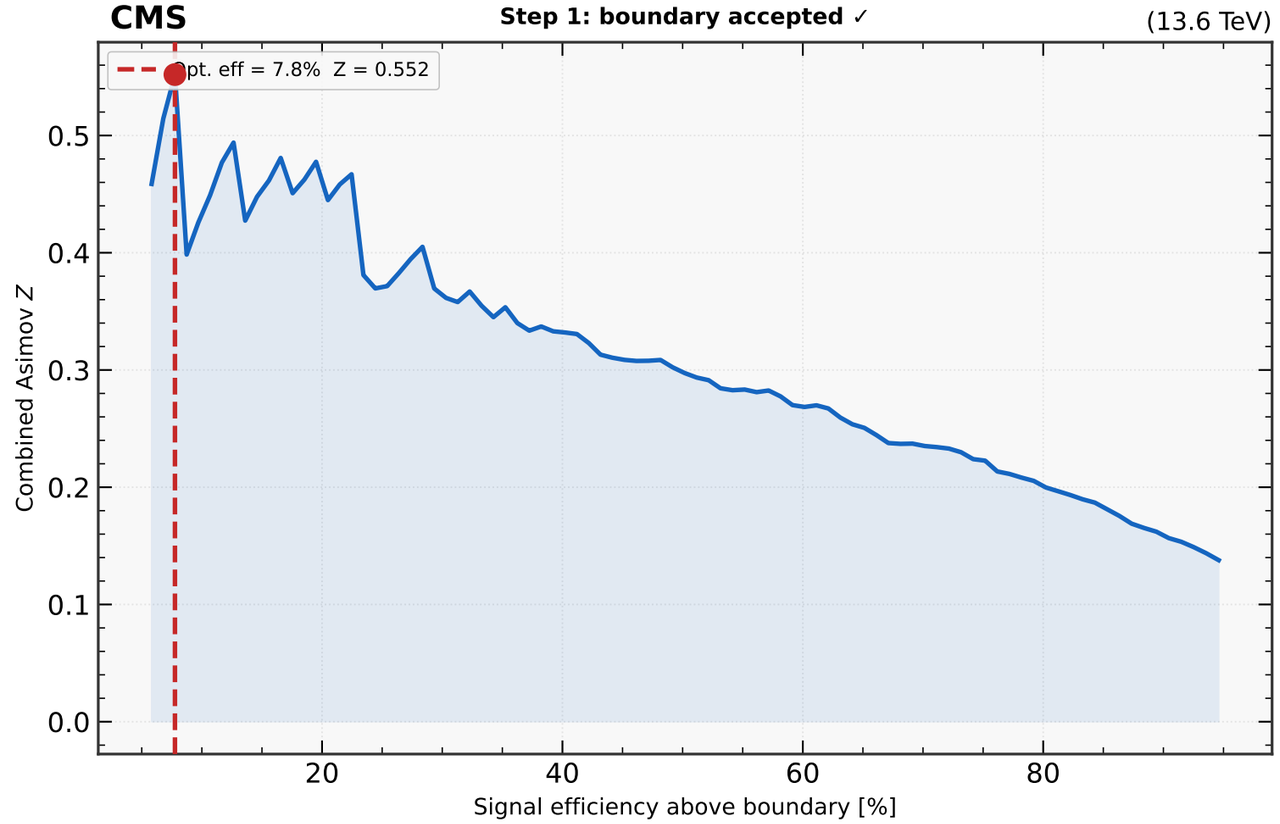
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.024	794.83 ± 12.01	367	0.00
[0.025, 0.050)	0.046	551.25 ± 9.63	472	0.01
[0.050, 0.075)	0.054	375.63 ± 7.83	366	0.01
[0.075, 0.100)	0.056	276.86 ± 6.59	302	0.02
[0.100, 0.125)	0.057	221.74 ± 5.95	250	0.03
[0.125, 0.150)	0.058	188.59 ± 5.52	204	0.03
[0.150, 0.175)	0.057	171.46 ± 5.27	195	0.03
[0.175, 0.200)	0.058	136.74 ± 4.66	156	0.04
[0.200, 0.225)	0.057	121.99 ± 4.37	122	0.05
[0.225, 0.250)	0.060	107.93 ± 4.17	130	0.06
[0.250, 0.275)	0.059	95.28 ± 3.88	106	0.06
[0.275, 0.300)	0.061	86.95 ± 3.76	100	0.07
[0.300, 0.325)	0.063	80.16 ± 3.63	82	0.08
[0.325, 0.350)	0.065	78.20 ± 3.50	84	0.08
[0.350, 0.375)	0.068	71.61 ± 3.36	78	0.10
[0.375, 0.400)	0.070	66.90 ± 3.23	98	0.10
[0.400, 0.425)	0.072	64.61 ± 3.24	91	0.11
[0.425, 0.450)	0.075	59.11 ± 3.01	66	0.13
[0.450, 0.475)	0.078	57.91 ± 3.07	73	0.13
[0.475, 0.500)	0.082	47.94 ± 2.78	70	0.17

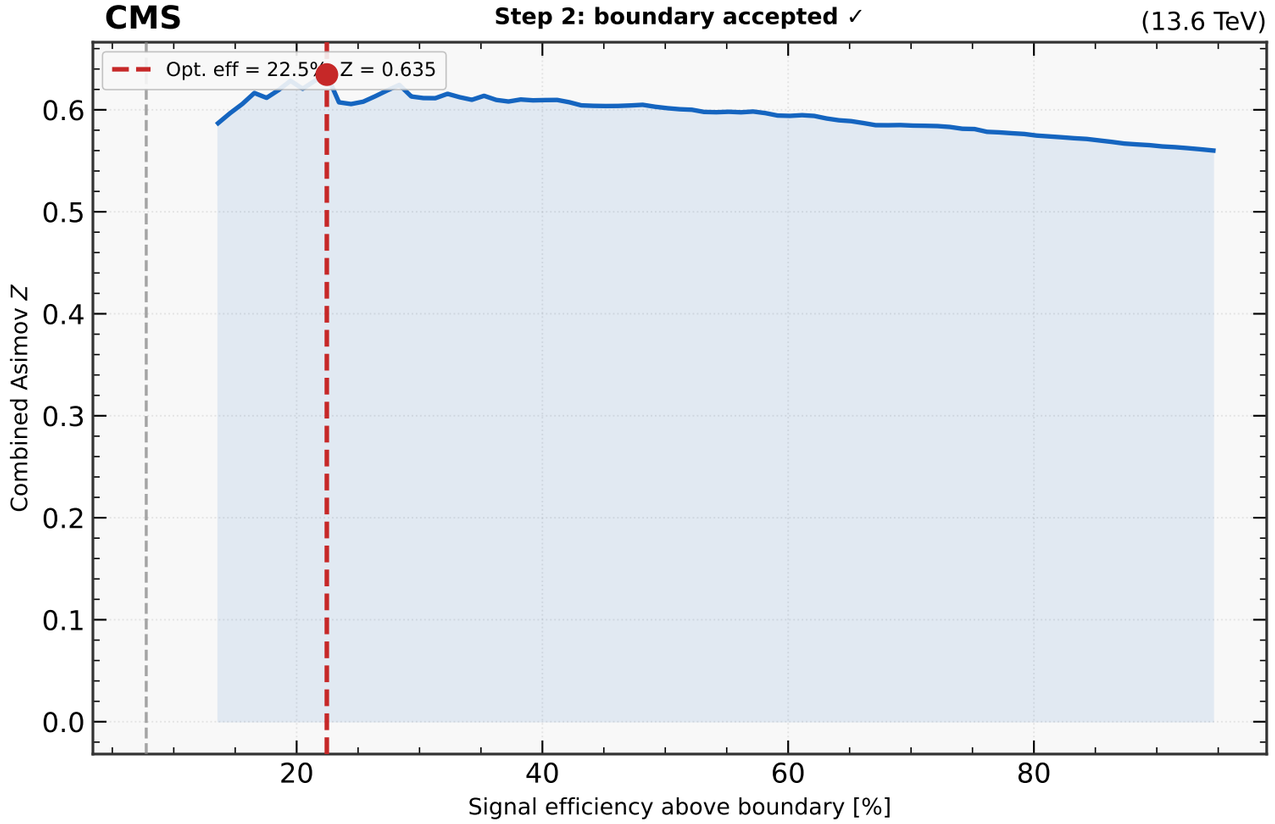
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

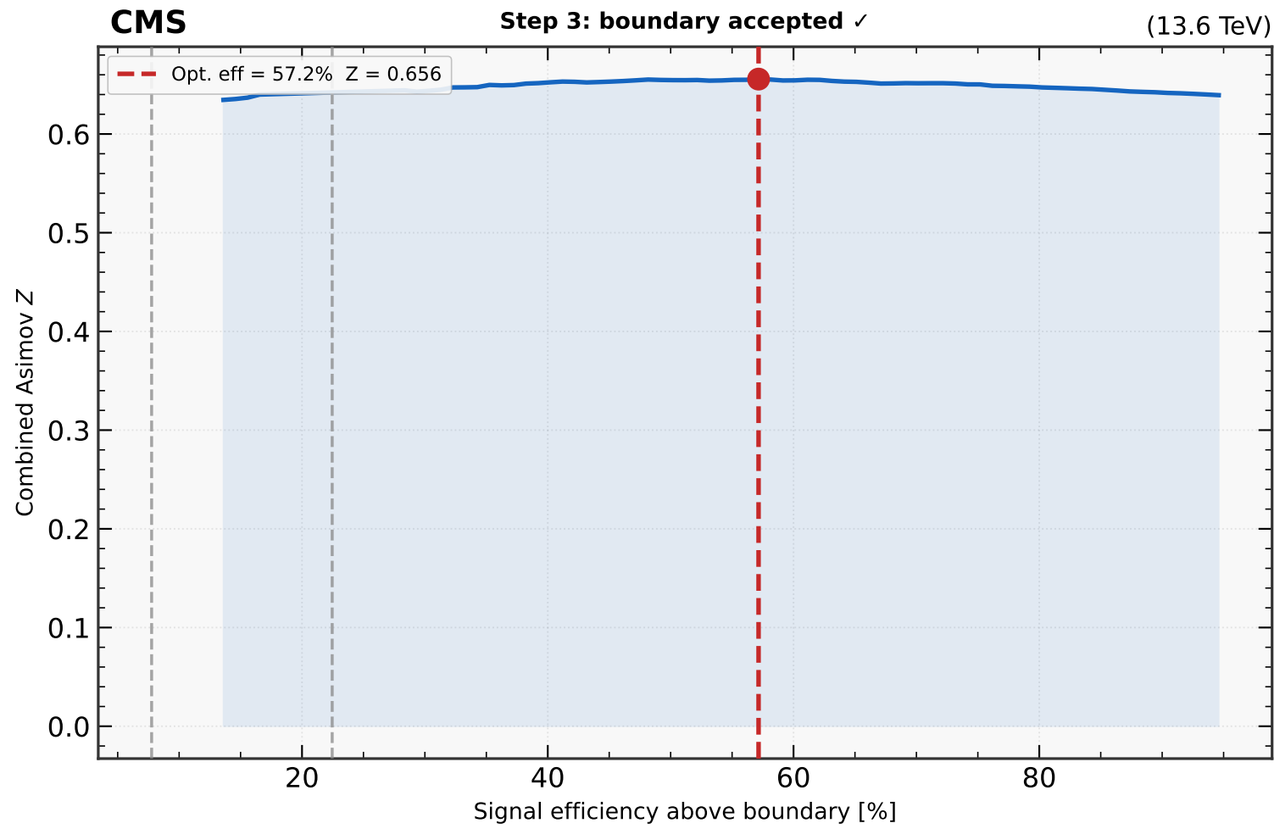
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.088	51.04 ± 2.82	55	0.17
[0.525, 0.550)	0.091	44.52 ± 2.66	61	0.20
[0.550, 0.575)	0.097	50.33 ± 2.81	69	0.19
[0.575, 0.600)	0.103	41.62 ± 2.55	59	0.25
[0.600, 0.625)	0.111	41.88 ± 2.55	56	0.26
[0.625, 0.650)	0.115	46.86 ± 2.77	42	0.24
[0.650, 0.675)	0.124	33.83 ± 2.27	57	0.37
[0.675, 0.700)	0.137	39.76 ± 2.50	45	0.34
[0.700, 0.725)	0.147	32.08 ± 2.17	49	0.46
[0.725, 0.750)	0.163	34.87 ± 2.29	51	0.47
[0.750, 0.775)	0.173	35.90 ± 2.40	57	0.48
[0.775, 0.800)	0.199	35.55 ± 2.32	54	0.56
[0.800, 0.825)	0.223	31.57 ± 2.17	57	0.70
[0.825, 0.850)	0.256	29.45 ± 2.06	51	0.86
[0.850, 0.875)	0.299	31.72 ± 2.22	39	0.94
[0.875, 0.900)	0.360	31.06 ± 2.17	41	1.15
[0.900, 0.925)	0.453	27.55 ± 2.01	52	1.62
[0.925, 0.950)	0.613	28.07 ± 2.05	42	2.14
[0.950, 0.975)	0.936	23.83 ± 1.82	40	3.78
[0.975, 1.000)	1.480	9.20 ± 1.01	25	13.86


Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.6556
0.091
0.552
0.635
0.656
Combined Asimov Z [HWHM window]
0.1
0.2
0.3
0.4
0.5
0.6
1
2
3
4
Number of BDT categories

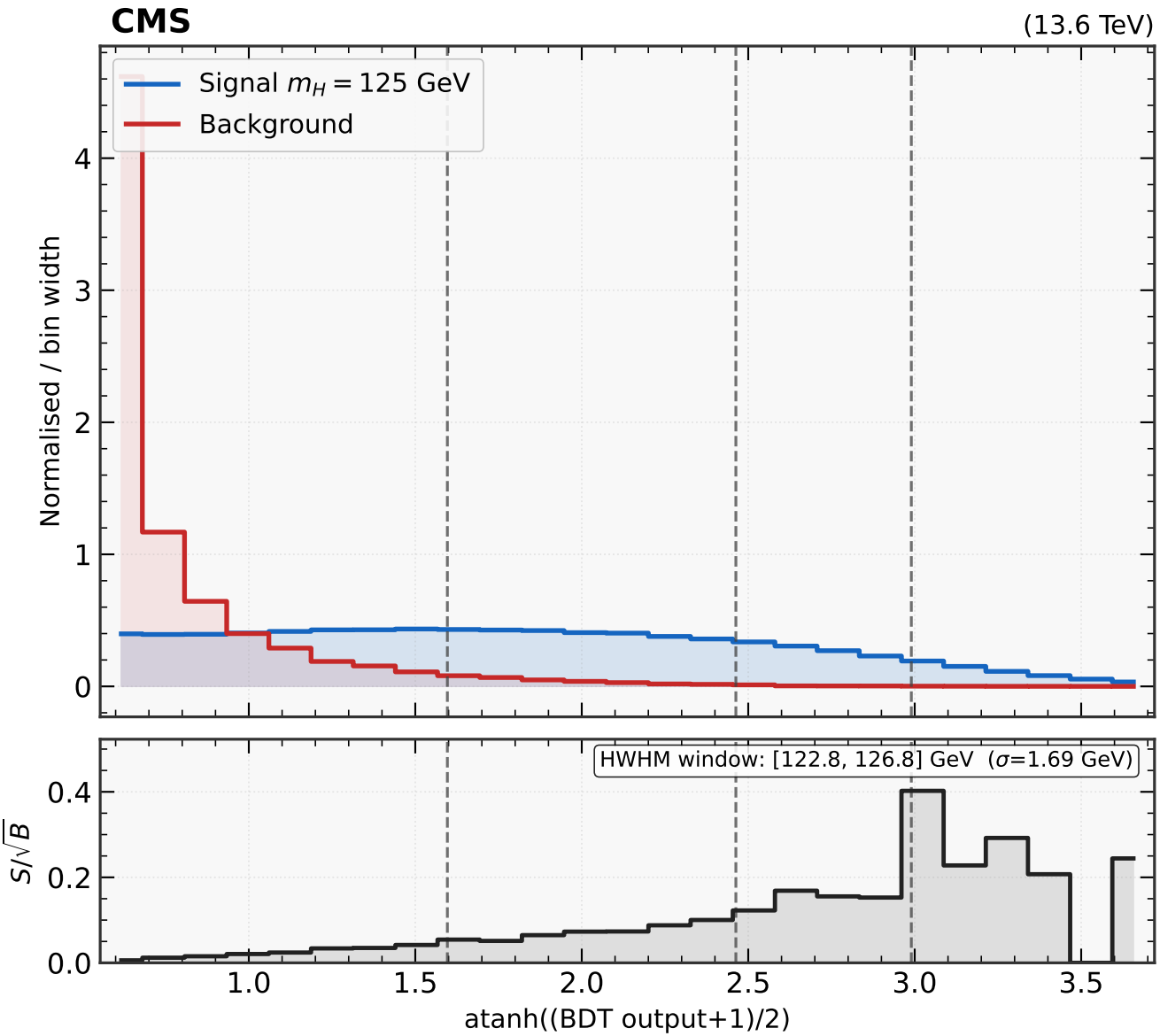
Iterative category boundary optimisation (AN Fig. 32 analog)

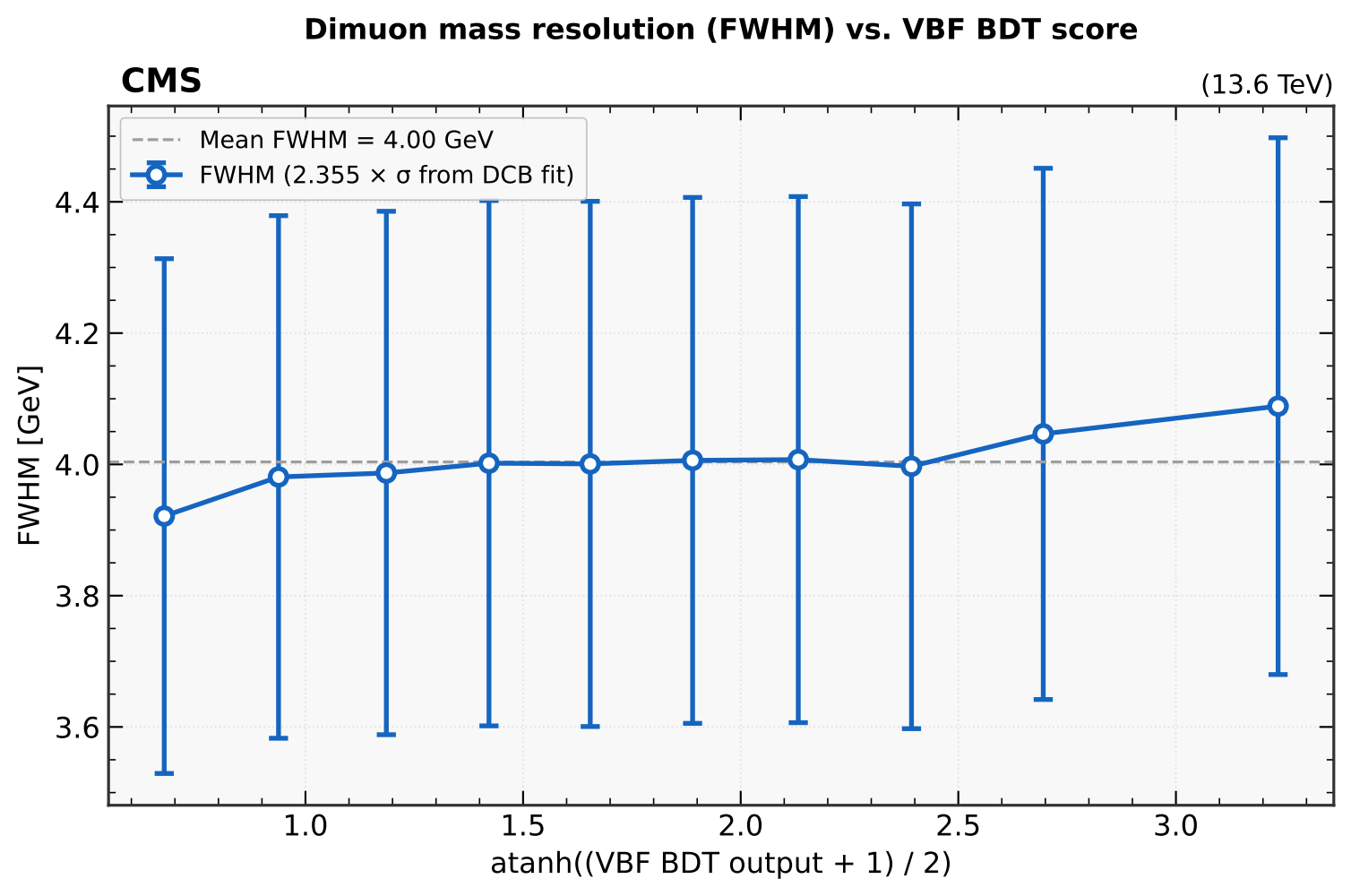


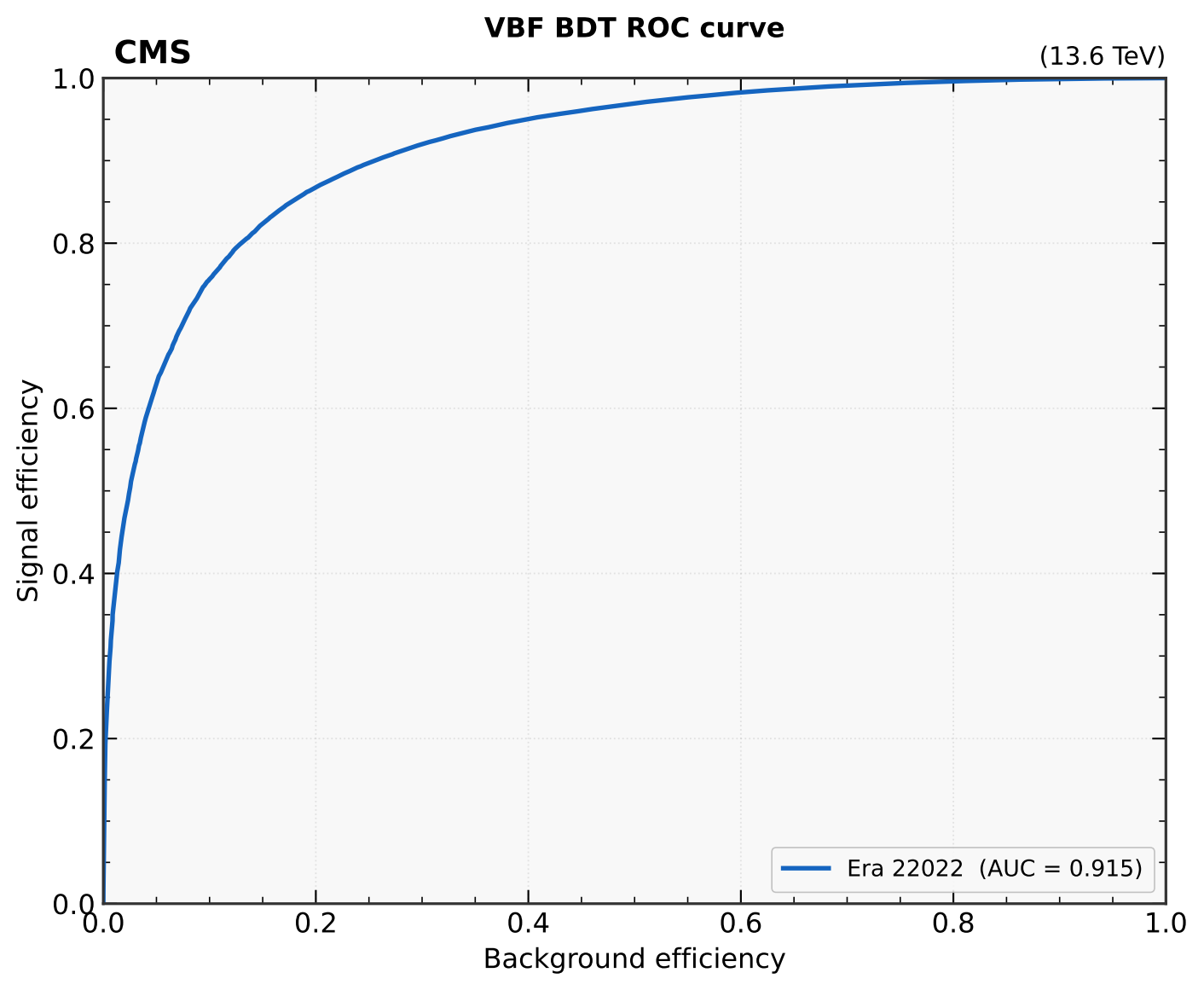


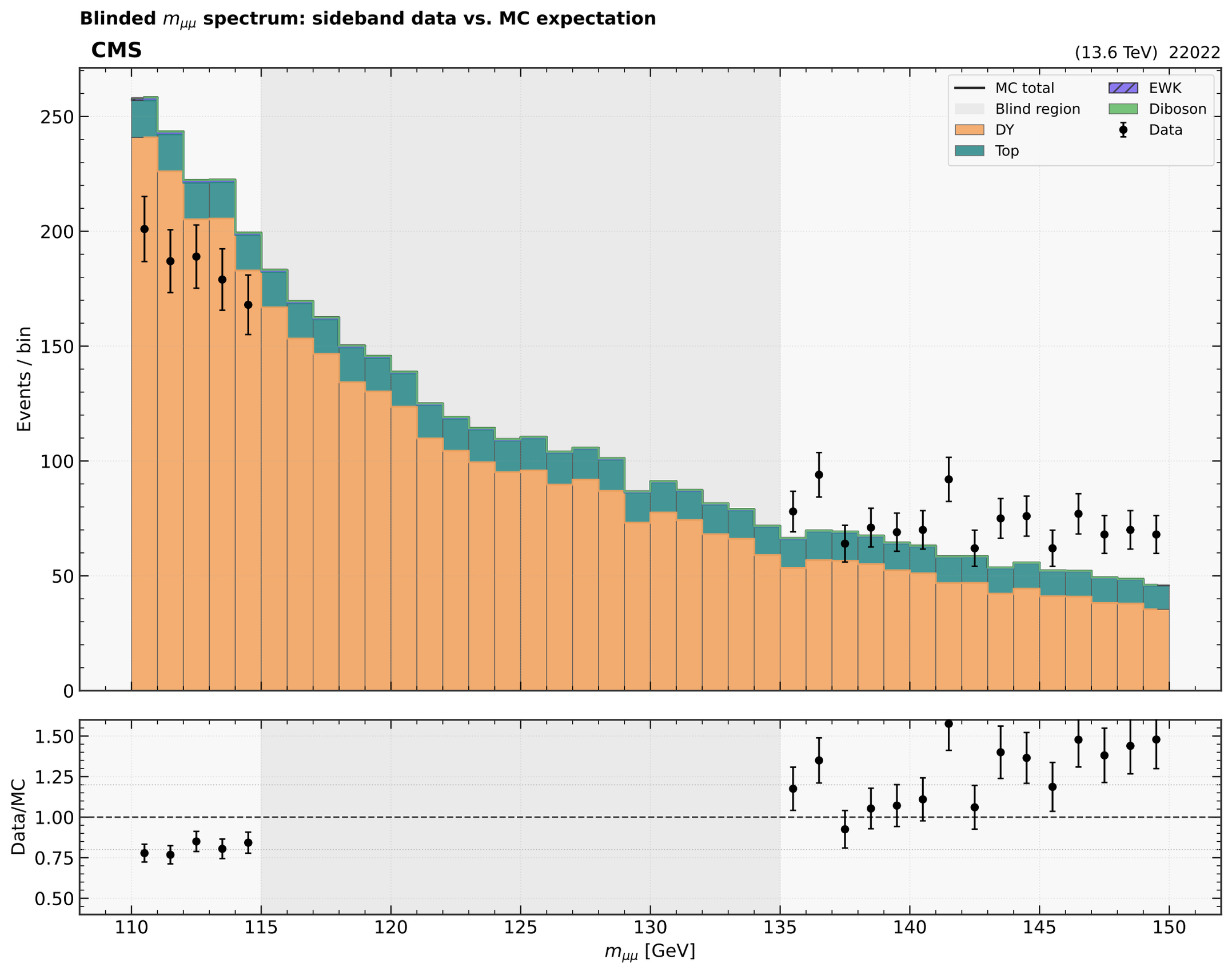


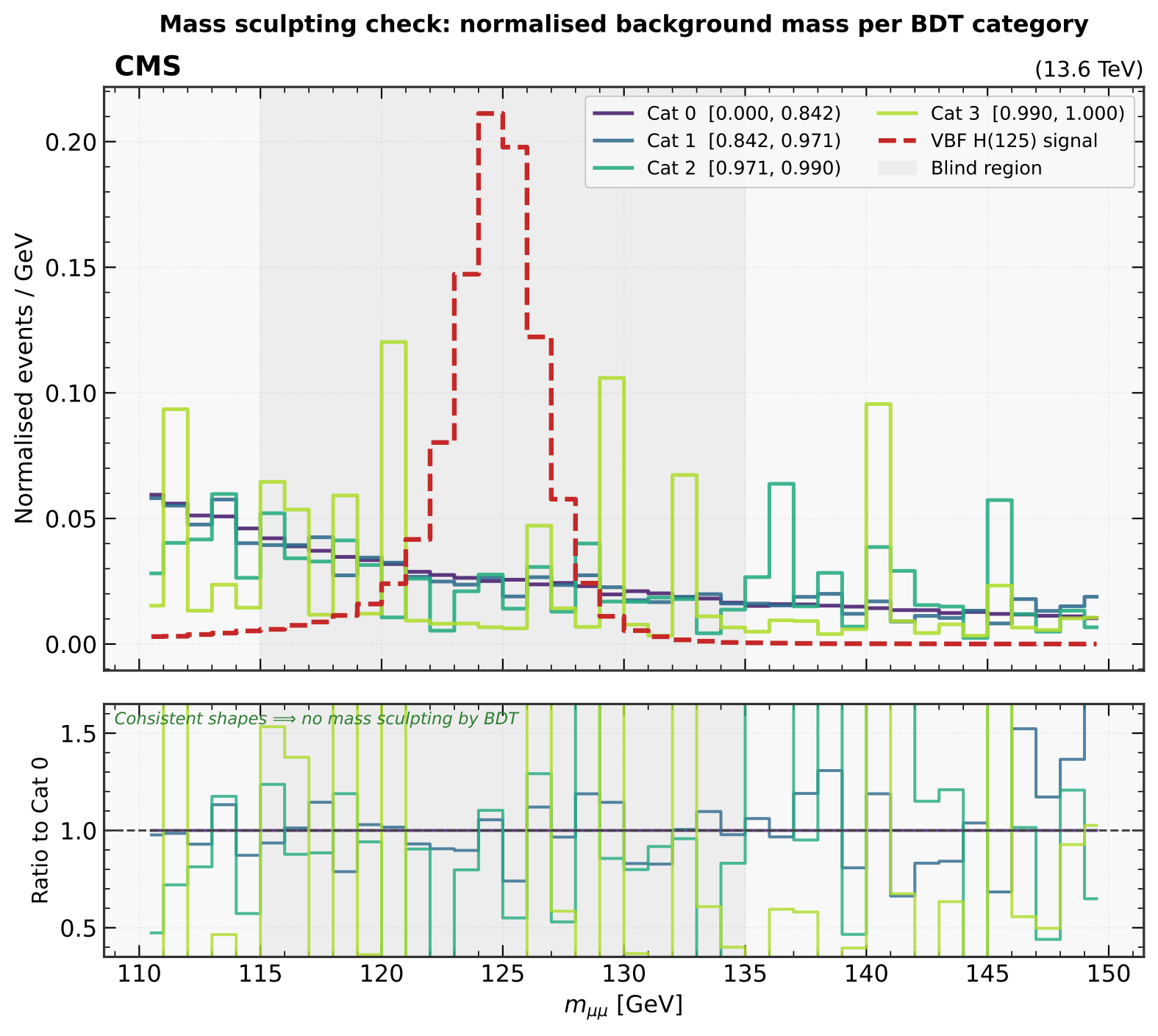
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window




Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 4.00 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

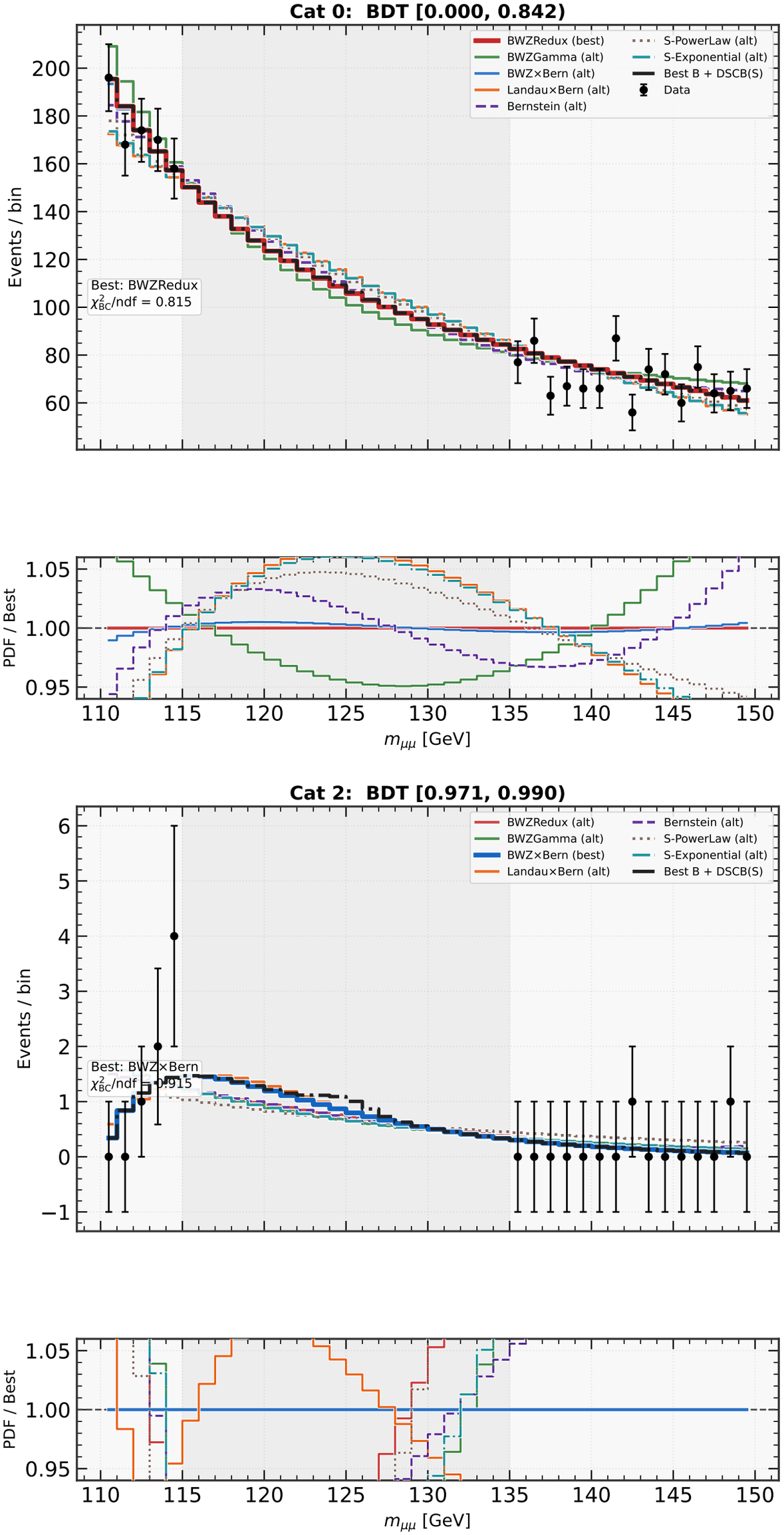

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 22022 (AUC = 0.915)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 22022
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

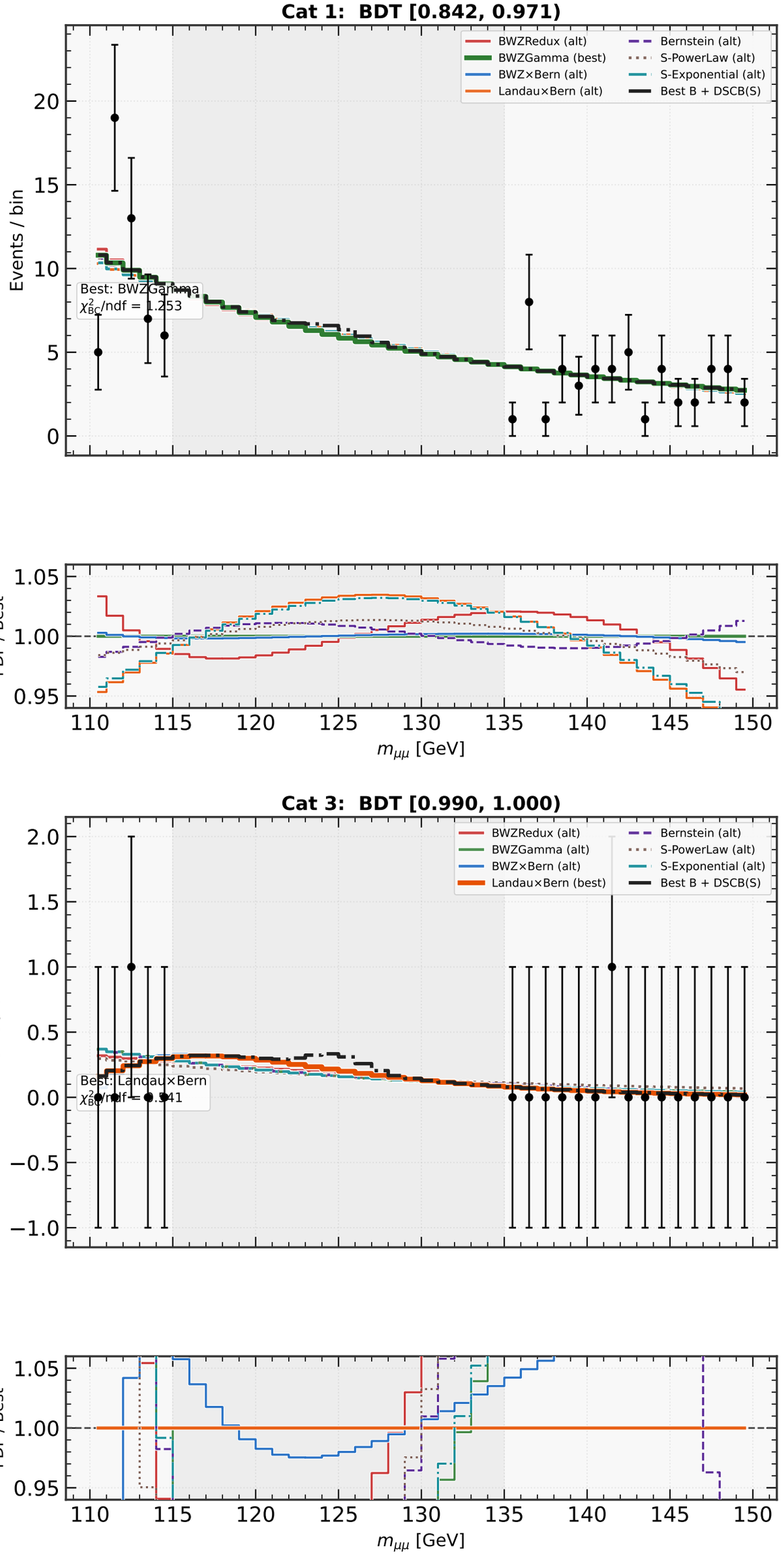

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.842)
Cat 1 [0.842, 0.971)
Cat 2 [0.971, 0.990)
Cat 3 [0.990, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.20
0.15
0.10
0.05
0.00
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
1.5
1.0
0.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

Cat 0: BDT [0.000, 0.842)



Cat 1: BDT [0.842, 0.971)



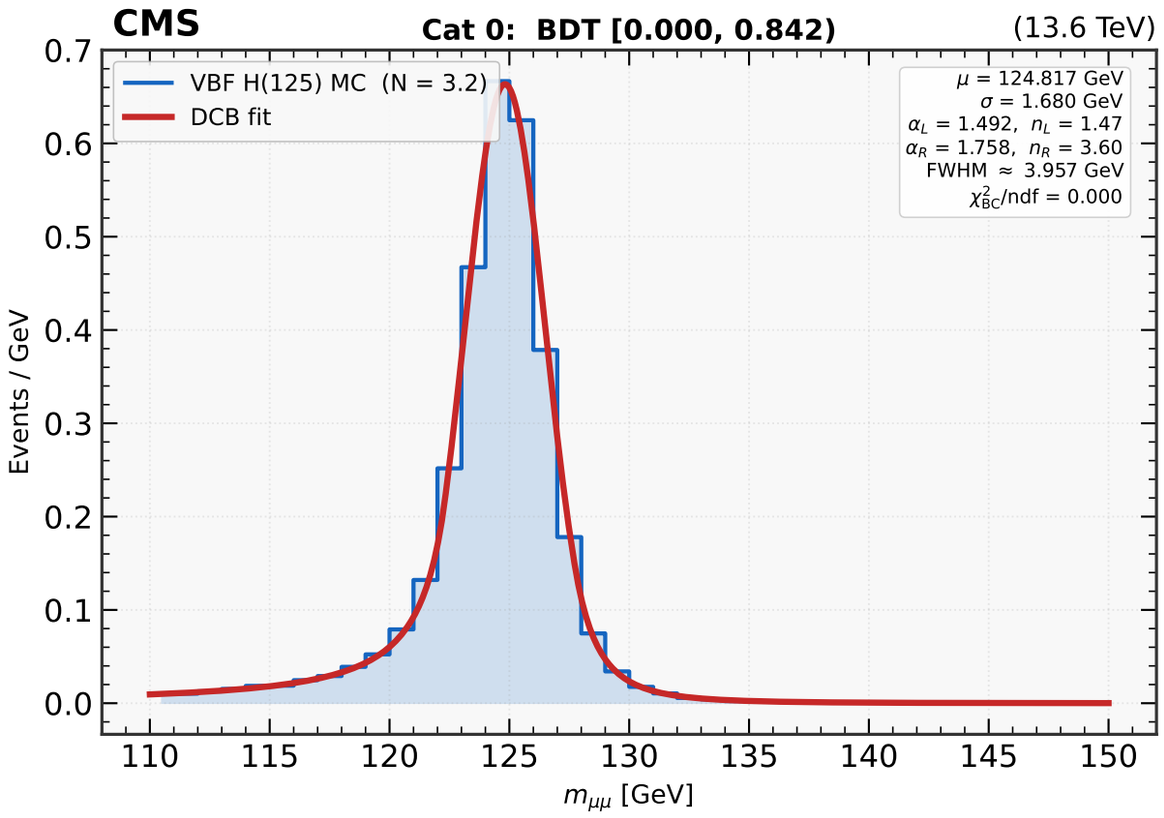
Cat 2: BDT [0.971, 0.990)

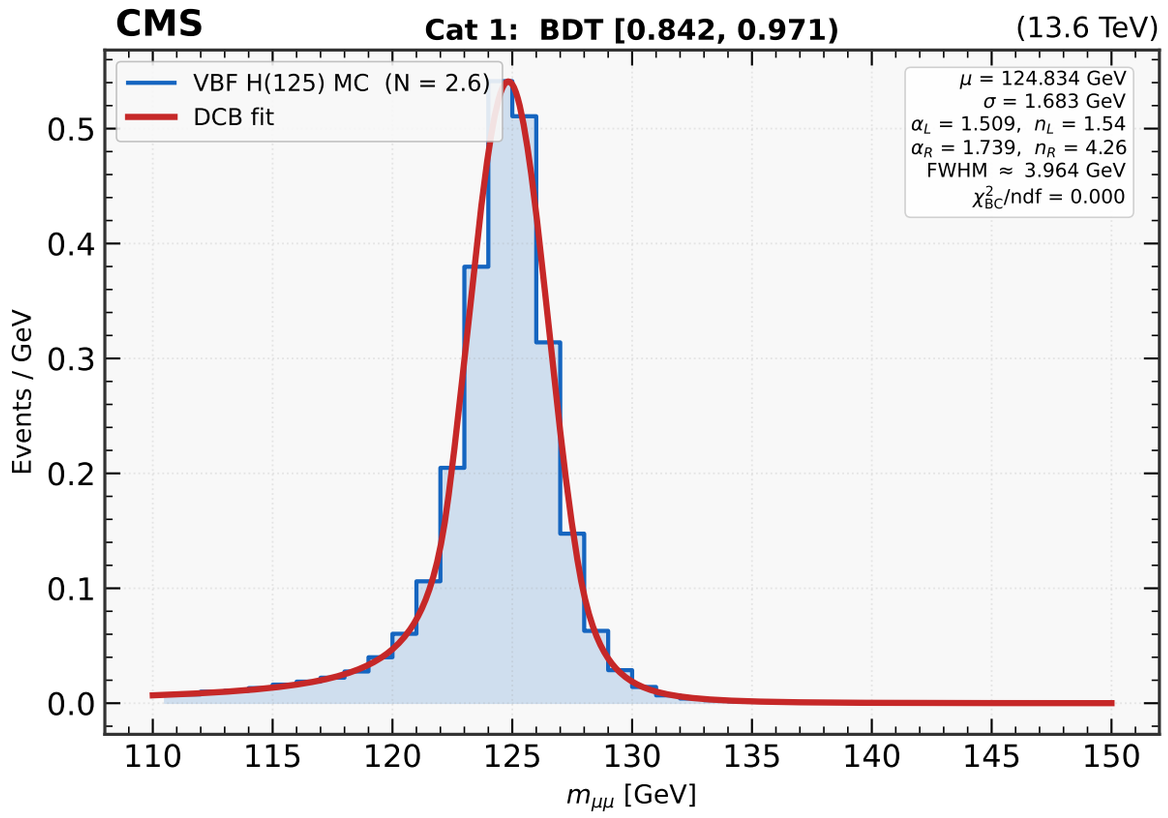
Cat 3: BDT [0.990, 1.000)

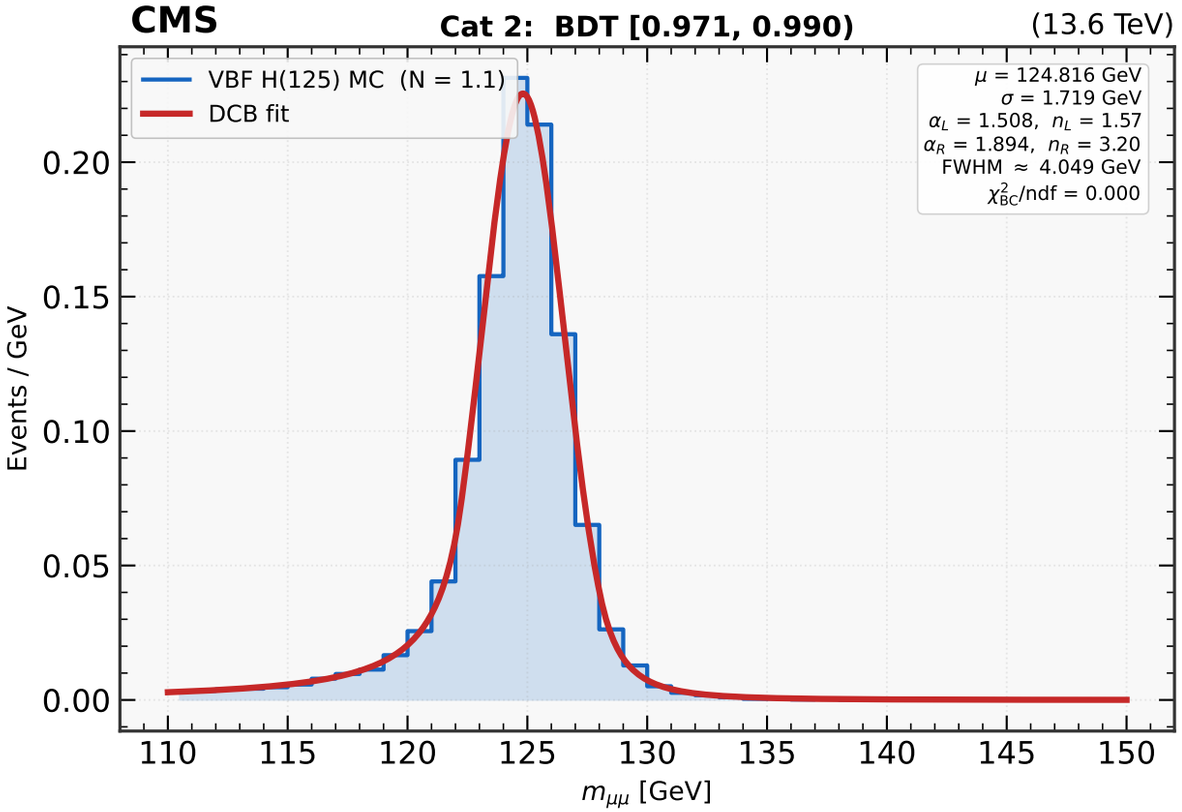
Per-category background model fit comparison (χ²_BC/ndf, ndf)

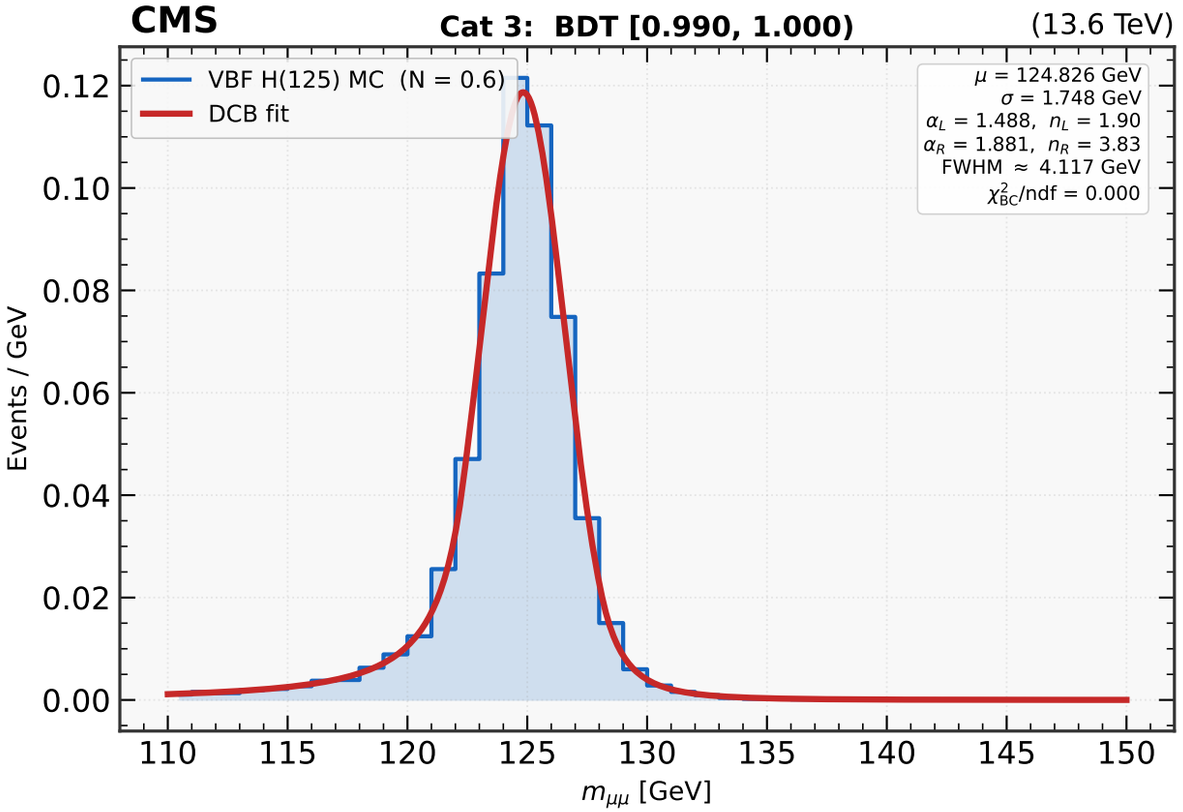
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.842)	BWZRedux	0.815	0.815 (37)	1.024 (37)	0.848 (35)	1.095 (37)	0.824 (35)	0.929 (37)	1.056 (37)
1	[0.842, 0.971)	BWZGamma	1.253	1.253 (37)	1.253 (37)	1.288 (36)	1.258 (37)	1.325 (35)	1.253 (37)	1.257 (37)
2	[0.971, 0.990)	BWZ×Bern	0.915	0.999 (37)	0.995 (37)	0.915 (36)	0.933 (37)	1.031 (35)	1.032 (37)	0.998 (37)
3	[0.990, 1.000)	Landau×Bern	0.541	0.552 (37)	0.553 (37)	0.549 (36)	0.541 (37)	0.580 (35)	0.560 (37)	0.553 (37)

VBF H(125) signal DCB extraction per BDT category



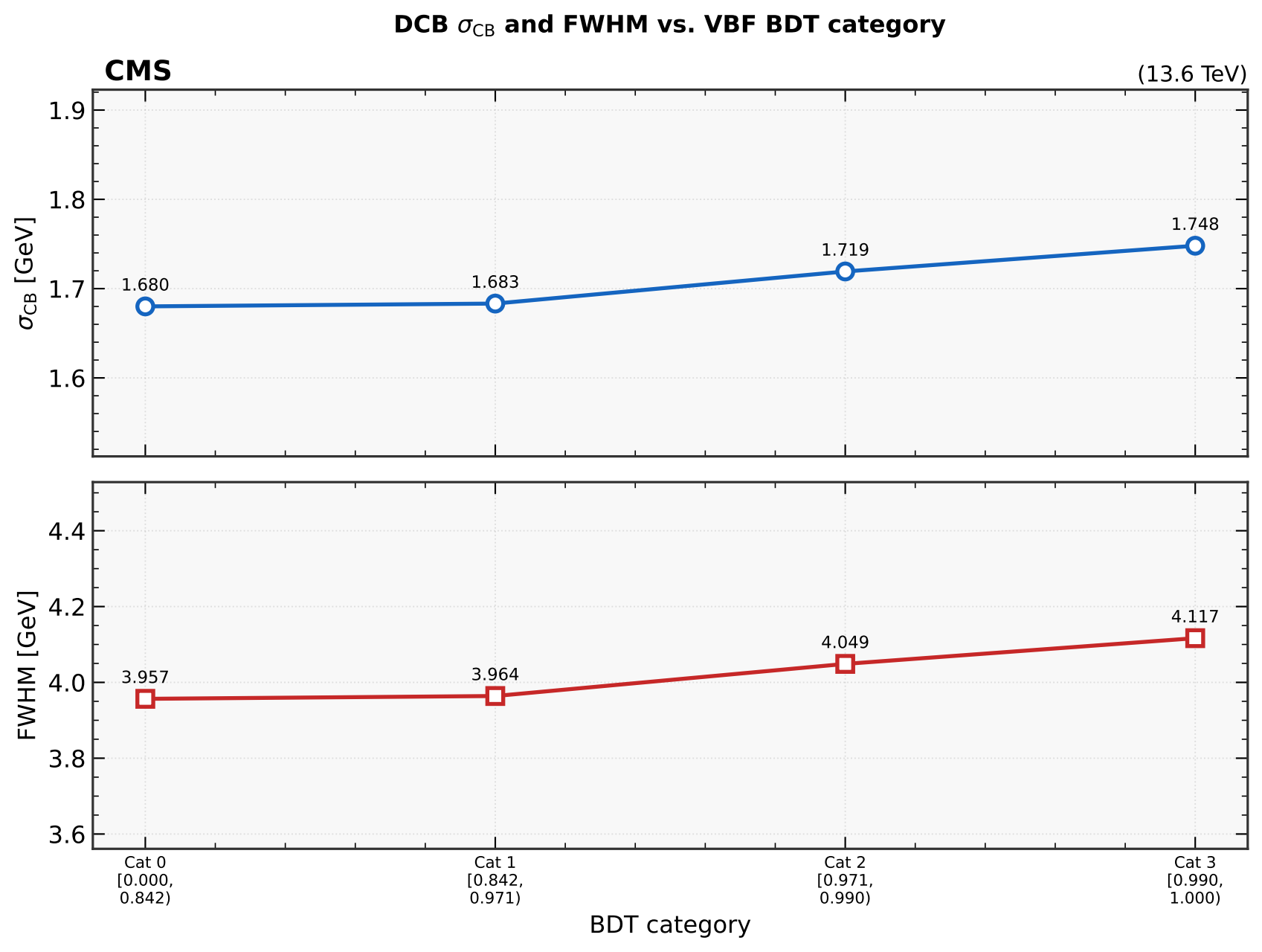


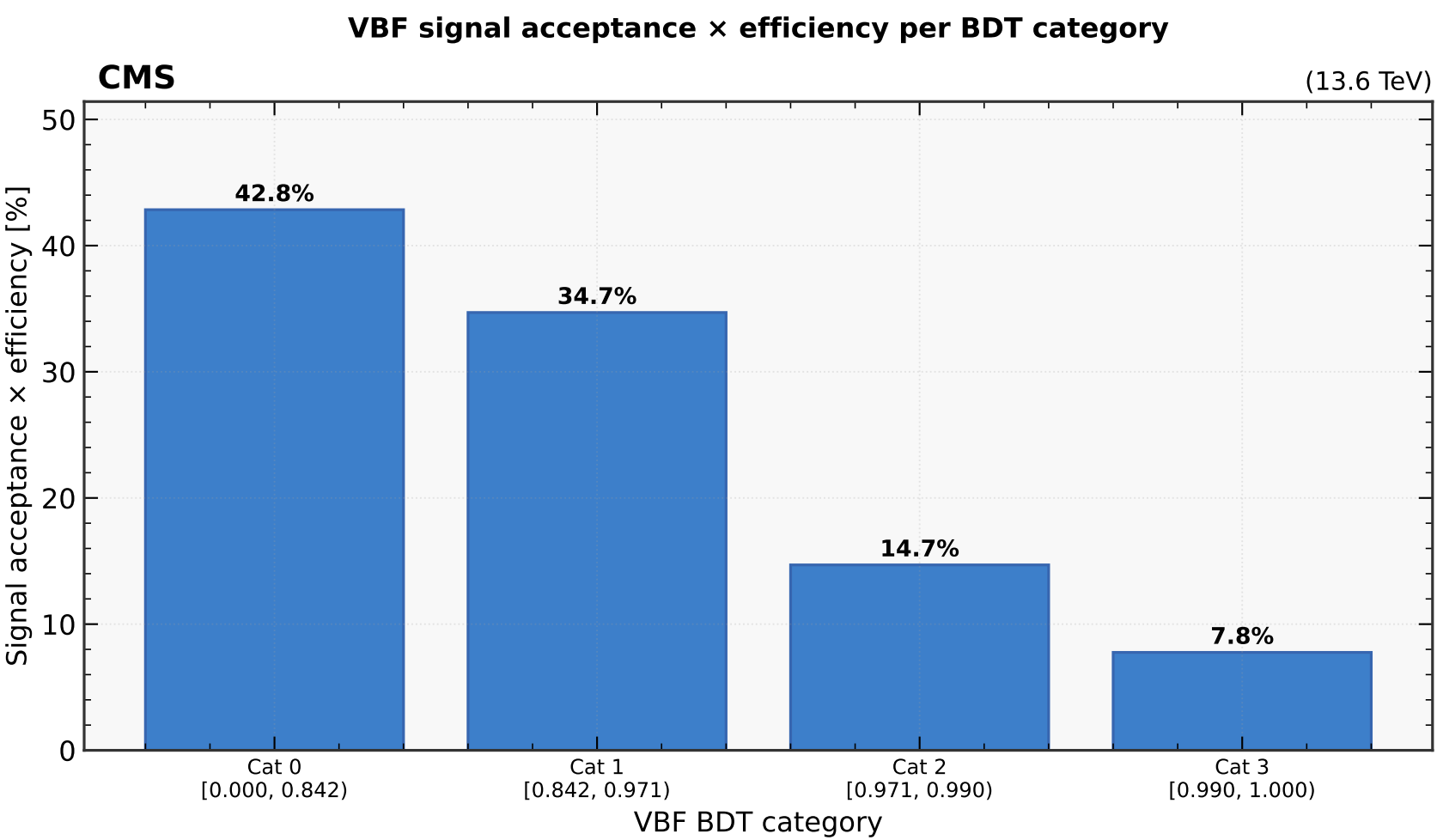




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.842)	3.17	124.8174	1.6801	1.4920	1.474	1.7578	3.603	3.957	0.039	0.000
1	[0.842, 0.971)	2.56	124.8341	1.6832	1.5085	1.539	1.7392	4.257	3.964	0.175	0.000
2	[0.971, 0.990)	1.09	124.8156	1.7192	1.5080	1.567	1.8943	3.198	4.049	0.319	0.000
3	[0.990, 1.000)	0.57	124.8263	1.7480	1.4876	1.903	1.8808	3.832	4.117	0.362	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.9
1.8
1.7
1.6
1.680
1.683
1.719
1.748
FWHM [GeV]
4.4
4.2
4.0
3.8
3.6
3.957
3.964
4.049
4.117
Cat 0 [0.000, 0.842)
Cat 1 [0.842, 0.971)
Cat 2 [0.971, 0.990)
Cat 3 [0.990, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
42.8%
34.7%
14.7%
7.8%
Cat 0
[0.000, 0.842)
Cat 1
[0.842, 0.971)
Cat 2
[0.971, 0.990)
Cat 3
[0.990, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.842)	3.17	29462.63	2.132	2923.774	[122.8, 126.8]	0.039	0.039
1	[0.842, 0.971)	2.56	1030.57	1.745	99.212	[122.9, 126.8]	0.175	0.175
2	[0.971, 0.990)	1.09	75.11	0.748	5.267	[122.8, 126.8]	0.326	0.319
3	[0.990, 1.000)	0.57	15.65	0.399	1.083	[122.8, 126.9]	0.383	0.362

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.842)	1910	13646.45 ± 124.96	0.140	29462.63 ± 184.49	0.0394	0.0394	2923.774
1	[0.842, 0.971)	99	489.26 ± 22.79	0.202	1030.57 ± 32.79	0.1752	0.1746	99.212
2	[0.971, 0.990)	9	39.94 ± 6.14	0.225	75.11 ± 8.21	0.3259	0.3186	5.267
3	[0.990, 1.000)	2	5.79 ± 1.92	0.345	15.65 ± 3.42	0.3829	0.3625	1.083
Combined	all						0.5147	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.842)	25747.89	49.23	3601.71	63.80	29462.63	34035.55
1	[0.842, 0.971)	905.54	32.45	88.96	3.63	1030.57	1075.26
2	[0.971, 0.990)	55.68	12.25	6.91	0.27	75.11	67.39
3	[0.990, 1.000)	9.82	4.91	0.89	0.03	15.65	11.69